



# Sai Krishna Sangeetha

## Full-Stack Developer

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### PROFESSIONAL SUMMARY

- Full-Stack Engineer with 5+ years of experience designing and delivering production SaaS platforms, distributed systems, observability products, and data-intensive applications across startup and enterprise environments.
- Strong hands-on engineering depth across React, TypeScript, Python, Django, FastAPI, REST/GraphQL APIs, PostgreSQL, MySQL, Redis, Kafka, Elasticsearch, Docker, Kubernetes, and AWS.
- Built cloud-native observability systems for logs, metrics, traces, service topology, alerting, anomaly detection, and root-cause analysis using Open Telemetry, Kafka, Logstash, Fluent Bit, Prometheus, Elasticsearch, Temporal, and LLM-assisted workflows.
- Experienced in cross-functional collaboration with product, platform, operations, and business stakeholders; comfortable explaining complex technical tradeoffs, leading reviews, troubleshooting production issues, and supporting high-stakes releases.
- Strong focus on system reliability, performance, security, and developer experience, including OAuth2, SAML, JWT, RBAC, CI/CD automation, cloud infrastructure, query optimization, caching, and monitoring.

### CORE TECHNICAL SKILLS

**Frontend Engineering:** React.js, TypeScript, JavaScript, Redux, Zustand, React Router, React Flow, Tailwind CSS, SASS, Webpack, Responsive Design, Component Architecture

**Backend Engineering:** Python, Django, FastAPI, Golang, REST APIs, GraphQL, Microservices, Async Processing, Redis, Celery, API Design, Distributed Workflows

**Cloud and Platform:** AWS Lambda, API Gateway, EC2, S3, CloudFront, Cognito, Docker, Kubernetes, Pulumi, Infrastructure as Code, Serverless Architecture, Linux

**Observability and Reliability:** Open Telemetry, Metrics, Traces, Logs, Prometheus, Elasticsearch, Kibana, Logstash, Fluent Bit, APM, Alerting, Telemetry Pipelines, RCA, Production Troubleshooting

**Data & Pipelines:** PostgreSQL, MySQL, Elasticsearch DSL, SQL Alchemy, Django ORM, Kafka, Temporal, Streaming Pipelines, Batch Processing, Query Optimization

**AI and Data Systems:** Anomaly Detection, Isolation Forest, Semantic Autoencoders, LLM Automation, Data Processing Pipelines, Feature Engineering, Llama, OpenAI, Amazon Bedrock, SageMaker

**Security and DevOps:** OAuth 2.0, JWT, SAML, SSO, RBAC, GitHub Actions, Jenkins, GitLab CI, CircleCI, AWS CodePipeline, Automated Deployments

**Ways of Working:** Technical Leadership, Architecture & Code Reviews, Agile/Scrum, Stakeholder Communication, Documentation, Unit Testing, TDD/BDD, Jira, Git, Postman

### EXPERIENCE

#### Associate Developer

Oct 2024 – Present

*Nova ITHUB Technology & Services LLC (Client: Ceburu Systems Inc.)*

– Pleasanton, CA – (Remote, FL)

- Architected and built a distributed observability platform for Logs, Metrics, Traces, NetFlow, Topology, Alerting, and Root-Cause Analysis using Open Telemetry, Kafka, Logstash, Fluent Bit, Prometheus, Elasticsearch, Django, React, and AWS.
- Designed Go-based edge telemetry agents for Linux and Windows to collect system logs, metrics, application traces, and network telemetry with secure enrollment, versioning, health monitoring, and cloud pipeline integration.
- Built high-throughput streaming ingestion pipelines processing millions of telemetry events per day with fault-tolerant consumers, replay capability, backpressure handling, scalable alerting workflows, and operational dashboards.
- Implemented OTEL-oriented instrumentation and pipeline patterns for distributed services, helping teams standardize how metrics, traces, and logs are collected, enriched, searched, and visualized.
- Developed AI-driven anomaly detection pipelines over S3-based telemetry and log data using Isolation Forest, semantic autoencoder models, rule-based detectors, and batch orchestration.
- Orchestrated AI-assisted RCA workflows using Temporal and LLaMA, forwarding anomaly signals with contextual logs to generate investigation summaries, likely causes, and recommended remediation actions.
- Built KQL-style log search and analytics APIs using Django, Elasticsearch DSL, pagination, aggregation strategies, and distributed query patterns for large telemetry datasets.
- Implemented dynamic service topology visualizations using React and React Flow, enabling teams to inspect service dependencies, performance signals, and incident context in real time.
- Built enterprise integrations including ServiceNow incident sync using REST APIs and webhooks, reducing manual operational effort by 40%.
- Designed multi-tenant RBAC with hierarchical module visibility, granular permissions, and secure access patterns for enterprise SaaS users.
- Developed an enterprise IPAM microservice for IPv4/IPv6 allocation, subnet discovery, reconciliation, and utilization tracking across network environments.
- Actively participated in architecture decisions, code reviews, sprint planning, release readiness, and production support for multiple product releases.

## Backend Software Developer

Mar 2024 – Sep 2024

*Transparency Wise Technology LLC (Now: AgWise LLC.)*

– St. Petersburg, FL

- Led full-stack development of a cloud-native AgTech platform using React, Django, MySQL, AWS, Docker, and Kubernetes for real-time agricultural analytics and nutrient recommendation workflows.
- Architected scalable ingestion and processing pipelines for soil samples, plant tissue tests, agronomic inputs, and lab data, enabling automated recommendations and real-time decision support.
- Designed serverless ETL workflows using AWS Lambda and Step Functions to automate email-based lab result ingestion, improving operational efficiency and increasing platform revenue by 30%.
- Migrated legacy REST endpoints to GraphQL APIs, enabling more flexible client-side data access and reducing average response latency by up to 85%.
- Built responsive React interfaces with Redux, React Router, SASS, and reusable component patterns to support data-rich analytics, forms, and workflow-heavy user experiences.
- Integrated Stripe for subscription payments and Mapbox for geospatial visualization, enabling location-aware analytics and commercial product packaging.
- Implemented CI/CD pipelines with AWS Code Pipeline, Docker, and Kubernetes to support repeatable deployments, auto-scaling, load balancing, and zero-downtime production releases.
- Improved application performance through SQL tuning, caching strategies, state management improvements, and frontend rendering optimization, reducing load times by 25%.
- Partnered with product and business stakeholders to refine requirements, explain technical tradeoffs, and deliver features aligned with customer workflows.

## Software Development Engineer

Jan 2019 – July 2022

*Freshworks Inc.,*

– Chennai, IN

- Engineered scalable backend services for a high-traffic multi-tenant HR SaaS platform using Python, Django, REST APIs, Redis, PostgreSQL/MySQL, AWS, and microservice architecture.
- Designed and deployed 45+ serverless APIs using AWS Lambda and API Gateway, enabling elastic scaling and high availability for global SaaS workloads.
- Architected event-driven workflows using Kafka for candidate updates, onboarding events, notifications, reporting, and cross-service synchronization.
- Implemented fault-tolerant asynchronous background processing using Redis-backed job queues for bulk candidate imports, email notifications, workflow automation, and reporting jobs.
- Refactored monolithic services into modular microservices, improving maintainability, deployment flexibility, observability, and team delivery speed.
- Designed AWS infrastructure using EC2, S3, CloudFront, and related services to support 10x traffic spikes during peak usage periods.
- Improved application responsiveness by 40% through lazy loading, client-side caching, background processing, API optimization, and frontend rendering improvements.
- Designed data-intensive dashboard interfaces for HR analytics, optimizing client-side processing, incremental rendering, and asynchronous API interactions for large datasets.
- Collaborated with product managers, QA, customer-facing teams, and support stakeholders to investigate issues, clarify requirements, and ship reliable customer-impacting features.

## SYSTEM DESIGN & ARCHITECTURE

- Distributed telemetry ingestion and observability pipelines for logs, metrics, traces, alerts, and service topology.
- Cloud-native SaaS architecture using React, Python, AWS, Docker, Kubernetes, serverless APIs, CI/CD, and multi-tenant access control.
- Event-driven microservices and asynchronous workflows using Kafka, Redis queues, Temporal, and background processing.
- Data-intensive search and analytics platforms using Elasticsearch, SQL databases, caching, indexing, and query optimization.
- AI-assisted operational workflows for anomaly detection, incident investigation, and root-cause analysis.

## EDUCATION

○ University of Central Florida – Masters in Computer Science

August 2022 – May 2024

○ SASTRA University – Bachelors in Information & Communication Technology

June 2015 – May 2019

## PROJECTS

### ○ GPT-Powered Chatbot

- Developed a **full-stack AI chatbot platform** with a React-based conversational interface and a FastAPI backend handling prompt orchestration and response streaming.
- Integrated **OpenAI GPT models** to generate context-aware responses, implementing conversation history management and prompt structuring for improved response relevance.
- Implemented **Redis-based caching and session management** to reduce response latency and support concurrent user interactions.

- Designed scalable backend services with **PostgreSQL persistence and asynchronous request handling** to support high-volume chat interactions.
- **Online Workers Hiring System**
  - Designed and implemented a **web-based workforce management platform** supporting user onboarding, job postings, task delegation, and workflow tracking.
  - Built modular frontend components and dynamic forms to support **real-time job assignment and worker task management workflows**.
  - Implemented secure **authentication and role-based user management**, enabling administrators to manage workforce assignments and operational workflows efficiently.
- **Deep Neural Network Compression**
  - Developed a **model compression framework for convolutional neural networks**, introducing a pruning strategy guided by explainable AI metrics and transfer learning techniques.
  - Reduced model size by **90% while maintaining 83% accuracy**, enabling deployment of deep learning models on **resource-constrained edge devices**.
  - Evaluated pruning strategies across multiple CNN architectures to identify optimal trade-offs between **model performance and computational efficiency**.
- **Abusive Comments Identification**
  - Built a **multilingual abusive content detection model** using BERT architecture trained on the MURIL dataset for Indian languages.
  - Designed preprocessing pipelines for multilingual text normalization and tokenization to improve model accuracy.
  - Achieved **81% classification accuracy**, enabling automated moderation of abusive content across multilingual social media datasets.
- **Cache Simulator**
  - Developed **configurable cache simulator** to analyse L1 cache performance across varying sizes (1KB–1MB) and associativity levels.
  - Implemented multiple cache replacement policies and analysed their impact on **cache miss rate and average access time (AAT)**.
  - Identified **Optimal Replacement policy** as the most efficient configuration for a 32KB cache, reducing miss rate by **45%** and improving access time by **30%**.

## PUBLICATIONS

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- **Various Dynamic Programming Methods to Solve Optimization Problems:** This paper explores multiple **optimization problems** and their algorithmic solutions. It addresses **the Longest Common Subsequence problem** and **Floyd-Warshall algorithm** while highlighting dynamic programming's essential elements: **optimal substructure** and **overlapping subproblems**.

## CERTIFICATIONS

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- [2024] **Angular** certified from Hackerrank.
- [2023] AWS Certified **Solutions Architect-Associate**.
- [2023] **Responsive Web Design** certified from freecodecamp.org.
- [2021] **Agile Project Management; Agile Project Delivery** certified from Udemy.
- [2020] **Java(Intermediate)** certified from Hackerrank.

## AWARDS & ACHIEVEMENTS

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- Awarded **“Will it Make the Boat Go Faster”** at Freshworks Inc., for delivering impactful initiatives and team collaboration.
- Awarded **“Annual Sprint Champion-2020”** at Freshworks Inc., for outstanding contribution.
- Secured a place in the top **5** in the **Hackathon** at Freshworks Inc., showcasing strong problem-solving, collaboration skills.
- Secured **first place** in the **Business Idea Presentation** event of **E-Cell** at SASTRA University.